CS 224n Assignment #2: Word2Vec and Dependency Parsing

Due Date: April 18th, Thursday, 4:30 PM PST.
In this assignment, you will review the mathematics behind Word2Vec and build a neural dependency parser using PyTorch. For a review of the fundamentals of PyTorch, please check out the PyTorch review session on Canvas. In Part 1, you will explore the partial derivatives involved in training a Word2vec model using the naive softmax loss. In Part 2, you will learn about two general neural network techniques (Adam Optimization and Dropout). In Part 3, you will implement and train a dependency parser using the techniques from Part 2, before analyzing a few erroneous dependency parses.
If you are using LaTeX, you can use \ifans{} to type your solutions.
Please tag the questions correctly on Gradescope, otherwise the TAs will take points off if you don't tag questions.

1. Understanding word2vec (15 points)

Recall that the key insight behind word2vec is that *'a word is known by the company it keeps'*. Concretely, consider a 'center' word c surrounded before and after by a context of a certain length. We term words in this contextual window 'outside words' (O). For example, in Figure 1, the context window length is 2, the center word c is 'banking', and the outside words are 'turning', 'into', 'crises', and 'as':

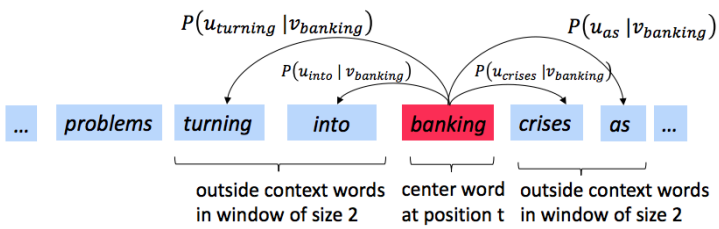


图 1: The word2vec skip-gram prediction model with window size 2

Skip-gram word2vec aims to learn the probability distribution $P(O|C)$. Specifically, given a specific word o and a specific word c, we want to predict $P(O = o|C = c)$: the probability that word o is an 'outside' word for c (i.e., that it falls within the contextual window of c). We model this probability by taking the softmax function over a series of vector dot-products:

$$P(O = o \mid C = c) = \frac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \tag{1}$$

For each word, we learn vectors u and v, where $\mathbf{u}_o$ is the 'outside' vector representing outside word o, and $\mathbf{v}_c$ is the 'center' vector representing center word c. We store these parameters in two matrices, $\mathbf{U}$ and $\mathbf{V}$. The columns of $\mathbf{U}$ are all the 'outside' vectors $\mathbf{u}_w$; the columns of $\mathbf{V}$ are all of the 'center' vectors $\mathbf{v}_w$. Both $\mathbf{U}$ and $\mathbf{V}$ contain a vector for every $w \in$ Vocabulary.[1]

[1] Assume that every word in our vocabulary is matched to an integer number k. Bolded lowercase letters represent vectors. $\mathbf{u}_k$ is both the kth column of $\mathbf{U}$ and the 'outside' word vector for the word indexed by k. $\mathbf{v}_k$ is both the kth column of $\mathbf{V}$ and the 'center' word vector for the word indexed by k. **In order to simplify notation we shall interchangeably use k to refer to word k and the index of word k.**

Recall from lectures that, for a single pair of words c and o, the loss is given by:

$$\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U}) = -\log P(O = o|C = c). \tag{2}$$

We can view this loss as the cross-entropy[2] between the true distribution $\mathbf{y}$ and the predicted distribution $\hat{\mathbf{y}}$, for a particular center word c and a particular outside word o. Here, both $\mathbf{y}$ and $\hat{\mathbf{y}}$ are vectors with length equal to the number of words in the vocabulary. Furthermore, the kth entry in these vectors indicates the conditional probability of the kth word being an 'outside word' for the given c. The true empirical distribution $\mathbf{y}$ is a one-hot vector with a 1 for the true outside word o, and 0 everywhere else, for this particular example of center word c and outside word o.[3] The predicted distribution $\hat{\mathbf{y}}$ is the probability distribution $P(O|C = c)$ given by our model in equation (1).

Note: Throughout this homework, when computing derivatives, please use the method reviewed during the lecture (i.e. no Taylor Series Approximations).

[2]The **cross-entropy loss** between the true (discrete) probability distribution p and another distribution q is $-\sum_i p_i \log(q_i)$.

[3]Note that the true conditional probability distribution of context words for the entire training dataset would not be one-hot.

(a) (2 points) Prove that the naive-softmax loss (Equation 2) is the same as the cross-entropy loss between $\mathbf{y}$ and $\hat{\mathbf{y}}$, i.e. (note that $\mathbf{y}$ (true distribution), $\hat{\mathbf{y}}$ (predicted distribution) are vectors and $\hat{\mathbf{y}}_o$ is a scalar):

$$-\sum_{w \in \text{Vocab}} \mathbf{y}_w \log(\hat{\mathbf{y}}_w) = -\log(\hat{\mathbf{y}}_o). \tag{3}$$

Your answer should be one line. You may describe your answer in words.

Solution: 该经验分布是一个大部分为 0 的向量，只有在 outside 的地方才为 1，也就是意味着：

$$-\sum_{w \in \text{Vocab}} \mathbf{y}_w \log(\hat{\mathbf{y}}_w) = -\sum_{w \neq o} 0 \cdot \log(\hat{\mathbf{y}}_w) - \sum_{w=o} 1 \cdot \log(\hat{\mathbf{y}}_w) = -\log(\hat{\mathbf{y}}_o).$$

(b) (6 points) i. Compute the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to $\mathbf{v}_c$. *Please write your answer in terms of* $\mathbf{y}$, $\hat{\mathbf{y}}$, $\mathbf{U}$, *and show your work to receive full credit.*

- **Note**: Your final answers for the partial derivative should follow the shape convention: the partial derivative of any function $f(x)$ with respect to x should have the **same shape** as x.[4]
- Please provide your answers for the partial derivative in vectorized form. For example, when we ask you to write your answers in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{U}$, you may not refer to specific elements of these terms in your final answer (such as $\mathbf{y}_1$, $\mathbf{y}_2$, ...).

ii. When is the gradient you computed equal to zero?
Hint: You may wish to review and use some introductory linear algebra concepts.

iii. The gradient you found is the difference between the two terms. Provide an interpretation of how each of these terms improves the word vector when this gradient is subtracted from the word vector v_c.

Solution:

i.

$$\begin{aligned}
\frac{\partial}{\partial \mathbf{v}_c} \mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U}) &= \frac{\partial}{\partial \mathbf{v}_c}[-\log(\exp(\mathbf{u}_o^\top \mathbf{v}_c)) + \log(\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c))] && (4)\\
&= \frac{\partial}{\partial \mathbf{v}_c}[-\mathbf{u}_o^\top \mathbf{v}_c + \log(\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c))] && (5)\\
&= -\mathbf{u}_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \frac{\partial}{\partial \mathbf{v}_c} \sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c) && (6)\\
&= -\mathbf{u}_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \cdot \sum_{w' \in \text{Vocab}} \exp(\mathbf{u}_{w'}^\top \mathbf{v}_c) \cdot \mathbf{u}_{w'} && (7)\\
&= -\mathbf{u}_o + \sum_{w' \in \text{Vocab}} \frac{\exp(\mathbf{u}_w^\top \mathbf{v}_c)}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \cdot \mathbf{u}_{w'} && (8)\\
&= -\mathbf{u}_o + \sum_{w' \in \text{Vocab}} \hat{\mathbf{y}}_{w'} \cdot \mathbf{u}_{w'} && (9)\\
&= -\mathbf{u}_o + \mathbf{U}\hat{\mathbf{y}} && (10)\\
&= -\mathbf{U}\mathbf{y} + \mathbf{U}\hat{\mathbf{y}} && (11)\\
&= \mathbf{U}(\hat{\mathbf{y}} - \mathbf{y}) && (12)
\end{aligned}$$

[4] This allows us to efficiently minimize a function using gradient descent without worrying about reshaping or dimension mismatching. While following the shape convention, we're guaranteed that $\theta := \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$ is a well-defined update rule.

ii. 根据求得的等式，当 $\hat{\mathbf{y}} = \mathbf{y}$ 时梯度为 0；根据矩阵的计算方式，当 $(\hat{\mathbf{y}} - \mathbf{y})$ 位于矩阵 $\mathbf{U}$ 的零空间时，梯度也为 0

iii. 根据定义的更新规则 $\theta = \theta - \alpha \frac{\partial \mathbf{J}(\theta)}{\partial \theta}$，也就是 $\mathbf{v}_c^{\text{new}} = \mathbf{v}_c^{\text{old}} - \alpha \mathbf{U}\hat{\mathbf{y}} + \alpha \mathbf{U}\mathbf{y}$，$\alpha$ 指的是学习率，$\mathbf{v}_c$ 加上 $\alpha \mathbf{U}\mathbf{y}$ 使得 $\mathbf{v}_c$ 向 $\mathbf{u}_o$ 的方向移动，也就是加强窗口内单词的向量对于单词 c 的向量的影响，而减去 $\alpha \mathbf{U}\hat{\mathbf{y}}$ 是使得 $\mathbf{v}_c$ 远离 $\mathbf{U}$ 中的其他向量

(c) (1 point) In many downstream applications using word embeddings, L2 normalized vectors (e.g. $\mathbf{u}/||\mathbf{u}||_2$ where $||\mathbf{u}||_2 = \sqrt{\sum_i u_i^2}$) are used instead of their raw forms (e.g. $\mathbf{u}$). Let's consider a hypothetical downstream task of binary classification of phrases as being positive or negative, where you decide the sign based on the sum of individual embeddings of the words. When would L2 normalization take away useful information for the downstream task? When would it not?

Hint: Consider the case where $\mathbf{u}_x = \alpha \mathbf{u}_y$ for some words $x \neq y$ and some scalar α. When α is positive, what will be the value of normalized $\mathbf{u}_x$ and normalized $\mathbf{u}_y$? How might $\mathbf{u}_x$ and $\mathbf{u}_y$ be related for such a normalization to affect or not affect the resulting classification?

Solution: 当向量的大小表示词频或者词的重要性时，L2 归一化会把所有向量的长度都标准化为 1，那么原始向量的长度信息就会丢失，使得性能低下。在某些任务中，仅仅只需要向量的方向时，词向量的方向比其大小要更加重要，L2 归一化把长度都标准化为 1，就可以去除向量长度对结果的影响，使模型更专注于词的含义等。

(d) (5 points) Compute the partial derivatives of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to each of the 'outside' word vectors, $\mathbf{u}_w$'s. There will be two cases: when $w = o$, the true 'outside' word vector, and $w \neq o$, for all other words. Please write your answer in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{v}_c$. In this subpart, you may use specific elements within these terms as well (such as $\mathbf{y}_1$, $\mathbf{y}_2$, ...). Note that $\mathbf{u}_w$ is a vector while $\mathbf{y}_1, \mathbf{y}_2, \ldots$ are scalars. Show your work to receive full credit.

Solution:

$$\frac{\partial}{\partial \mathbf{u}_w} \mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U}) = \frac{\partial}{\partial \mathbf{u}_w}[-\log \exp(\mathbf{u}_o^\top \mathbf{v}_c) + \log \sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)] \tag{13}$$

$$= -\frac{\partial \mathbf{u}_o^\top \mathbf{v}_c}{\partial \mathbf{u}_w} + \frac{1}{\sum_{w' \in \text{Vocab}} \exp(\mathbf{u}_{w'}^\top \mathbf{v}_c)} \cdot \exp(\mathbf{u}_w^\top \mathbf{v}_c) \cdot \mathbf{v}_c \tag{14}$$

$$= -\frac{\partial \mathbf{u}_o^\top \mathbf{v}_c}{\partial \mathbf{u}_w} + \hat{\mathbf{y}_w} \cdot \mathbf{v}_c \tag{15}$$

$$= \begin{cases} (\hat{\mathbf{y}_w} - 1)\mathbf{v}_c & \text{, if } w = o \\ \hat{\mathbf{y}_w}\mathbf{v}_c & \text{, otherwise} \end{cases} \tag{16}$$

(e) (1 point) Write down the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to $\mathbf{U}$. Please break down your answer in terms of the column vectors $\frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_1}, \frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_2}, \ldots, \frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_{|\text{Vocab}|}}$. No derivations are necessary, just an answer in the form of a matrix.

Solution:

$$\frac{\partial}{\partial \mathbf{U}} \mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U}) = \begin{bmatrix} \hat{\mathbf{y}}_1 \mathbf{v}_c & \hat{\mathbf{y}}_2 \mathbf{v}_c & \ldots & (\hat{\mathbf{y}}_o - 1)\mathbf{v}_c & \ldots & \hat{\mathbf{y}}_{|\text{Vocab}|} \mathbf{v}_c \end{bmatrix} \tag{17}$$

$$= (\hat{\mathbf{y}} - \mathbf{y})^\top \mathbf{v}_c \tag{18}$$

2. Machine Learning & Neural Networks (8 points)

(a) (4 points) Adam Optimizer

Recall the standard Stochastic Gradient Descent update rule:

$$\boldsymbol{\theta}_{t+1} \leftarrow \boldsymbol{\theta}_t - \alpha \nabla_{\boldsymbol{\theta}_t} J_{\text{minibatch}}(\boldsymbol{\theta}_t)$$

where $t+1$ is the current timestep, $\boldsymbol{\theta}$ is a vector containing all of the model parameters, ($\boldsymbol{\theta}_t$ is the model parameter at time step t, and $\boldsymbol{\theta}_{t+1}$ is the model parameter at time step $t+1$), J is the loss function, $\nabla_{\boldsymbol{\theta}} J_{\text{minibatch}}(\boldsymbol{\theta})$ is the gradient of the loss function with respect to the parameters on a minibatch of data, and α is the learning rate. Adam Optimization[5] uses a more sophisticated update rule with two additional steps.[6]

i. (2 points) First, Adam uses a trick called *momentum* by keeping track of $\mathbf{m}$, a rolling average of the gradients:

$$\begin{aligned}
\mathbf{m}_{t+1} &\leftarrow \beta_1 \mathbf{m}_t + (1-\beta_1) \nabla_{\boldsymbol{\theta}_t} J_{\text{minibatch}}(\boldsymbol{\theta}_t) \\
\boldsymbol{\theta}_{t+1} &\leftarrow \boldsymbol{\theta}_t - \alpha \mathbf{m}_{t+1}
\end{aligned}$$

where β_1 is a hyperparameter between 0 and 1 (often set to 0.9). Briefly explain in 2–4 sentences (you don't need to prove mathematically, just give an intuition) how using $\mathbf{m}$ stops the updates from varying as much and why this low variance may be helpful to learning, overall.

ii. (2 points) Adam extends the idea of *momentum* with the trick of *adaptive learning rates* by keeping track of $\mathbf{v}$, a rolling average of the magnitudes of the gradients:

$$\begin{aligned}
\mathbf{m}_{t+1} &\leftarrow \beta_1 \mathbf{m}_t + (1-\beta_1) \nabla_{\boldsymbol{\theta}_t} J_{\text{minibatch}}(\boldsymbol{\theta}_t) \\
\mathbf{v}_{t+1} &\leftarrow \beta_2 \mathbf{v}_t + (1-\beta_2)(\nabla_{\boldsymbol{\theta}_t} J_{\text{minibatch}}(\boldsymbol{\theta}_t) \odot \nabla_{\boldsymbol{\theta}_t} J_{\text{minibatch}}(\boldsymbol{\theta}_t)) \\
\boldsymbol{\theta}_{t+1} &\leftarrow \boldsymbol{\theta}_t - \alpha \mathbf{m}_{t+1} / \sqrt{\mathbf{v}_{t+1}}
\end{aligned}$$

where $\odot$ and $/$ denote elementwise multiplication and division (so $\mathbf{z} \odot \mathbf{z}$ is elementwise squaring) and β_2 is a hyperparameter between 0 and 1 (often set to 0.99). Since Adam divides the update by $\sqrt{\mathbf{v}}$, which of the model parameters will get larger updates? Why might this help with learning?

(b) (4 points) Dropout[7] is a regularization technique. During training, dropout randomly sets units in the hidden layer $\mathbf{h}$ to zero with probability p_{drop} (dropping different units each minibatch), and then multiplies $\mathbf{h}$ by a constant γ. We can write this as:

$$\mathbf{h}_{\text{drop}} = \gamma \mathbf{d} \odot \mathbf{h}$$

where $\mathbf{d} \in \{0,1\}^{D_h}$ (D_h is the size of $\mathbf{h}$) is a mask vector where each entry is 0 with probability p_{drop} and 1 with probability $(1-p_{\text{drop}})$. γ is chosen such that the expected value of $\mathbf{h}_{\text{drop}}$ is $\mathbf{h}$:

$$\mathbb{E}_{p_{\text{drop}}}[\mathbf{h}_{\text{drop}}]_i = h_i$$

for all $i \in \{1, \ldots, D_h\}$.

[5] Kingma and Ba, 2015, https://arxiv.org/pdf/1412.6980.pdf

[6] The actual Adam update uses a few additional tricks that are less important, but we won't worry about them here. If you want to learn more about it, you can take a look at: http://cs231n.github.io/neural-networks-3/#sgd

[7] Srivastava et al., 2014, https://www.cs.toronto.edu/~hinton/absps/JMLRdropout.pdf

i. (2 points) What must γ equal in terms of p_{drop}? Briefly justify your answer or show your math derivation using the equations given above.

ii. (2 points) Why should dropout be applied during training? Why should dropout **NOT** be applied during evaluation? **Hint:** it may help to look at the dropout paper linked.

3. Neural Transition-Based Dependency Parsing (54 points)

In this section, you'll be implementing a neural-network based dependency parser with the goal of maximizing performance on the UAS (Unlabeled Attachment Score) metric.

Before you begin, please follow the README to install all the needed dependencies for the assignment. We will be using PyTorch 2.1.2 from https://pytorch.org/get-started/locally/ with the CUDA option set to None, and the tqdm package – which produces progress bar visualizations throughout your training process. The official PyTorch website is a great resource that includes tutorials for understanding PyTorch's Tensor library and neural networks.

A dependency parser analyzes the grammatical structure of a sentence, establishing relationships between *head* words, and words which modify those heads. There are multiple types of dependency parsers, including transition-based parsers, graph-based parsers, and feature-based parsers. Your implementation will be a *transition-based* parser, which incrementally builds up a parse one step at a time. At every step it maintains a *partial parse*, which is represented as follows:

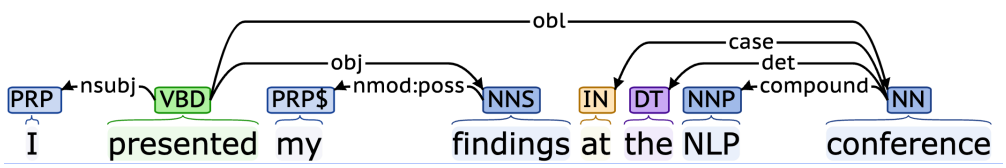
- A *stack* of words that are currently being processed.
- A *buffer* of words yet to be processed.
- A list of *dependencies* predicted by the parser.

Initially, the stack only contains ROOT, the dependencies list is empty, and the buffer contains all words of the sentence in order. At each step, the parser applies a *transition* to the partial parse until its buffer is empty and the stack size is 1. The following transitions can be applied:

- SHIFT: removes the first word from the buffer and pushes it onto the stack.
- LEFT-ARC: marks the second (second most recently added) item on the stack as a dependent of the first item and removes the second item from the stack, adding a *first_word* → *second_word* dependency to the dependency list.
- RIGHT-ARC: marks the first (most recently added) item on the stack as a dependent of the second item and removes the first item from the stack, adding a *second_word* → *first_word* dependency to the dependency list.

On each step, your parser will decide among the three transitions using a neural network classifier.

(a) (4 points) Go through the sequence of transitions needed for parsing the sentence *"I presented my findings at the NLP conference"*. The dependency tree for the sentence is shown below. At each step, give the configuration of the stack and buffer, as well as what transition was applied this step and what new dependency was added (if any). The first three steps are provided below as an example.



Stack	Buffer	New dependency	Transition
[ROOT]	[I, presented, my, findings, at, the, NLP, conference]		Initial Configuration
[ROOT, I]	[presented, my, findings, at, the, NLP, conference]		SHIFT
[ROOT, I, presented]	[my, findings, at, the, NLP, conference]		SHIFT
[ROOT, presented]	[my, findings, at, the, NLP, conference]	presented→I	LEFT-ARC

(b) (2 points) A sentence containing n words will be parsed in how many steps (in terms of n)? Briefly explain in 1–2 sentences why.

(c) (6 points) Implement the __init__ and parse_step functions in the PartialParse class in parser_transitions.py. This implements the transition mechanics your parser will use. You can run basic (non-exhaustive) tests by running python parser_transitions.py part_c.

(d) (8 points) Our network will predict which transition should be applied next to a partial parse. We could use it to parse a single sentence by applying predicted transitions until the parse is complete. However, neural networks run much more efficiently when making predictions about *batches* of data at a time (i.e., predicting the next transition for any different partial parses simultaneously). We can parse sentences in minibatches with the following algorithm.

Algorithm 1 Minibatch Dependency Parsing

Input: sentences, a list of sentences to be parsed and model, our model that makes parse decisions

Initialize partial_parses as a list of PartialParses, one for each sentence in sentences
Initialize unfinished_parses as a shallow copy of partial_parses
while unfinished_parses is not empty **do**
  Take the first batch_size parses in unfinished_parses as a minibatch
  Use the model to predict the next transition for each partial parse in the minibatch
  Perform a parse step on each partial parse in the minibatch with its predicted transition
  Remove the completed (empty buffer and stack of size 1) parses from unfinished_parses
end while

Return: The dependencies for each (now completed) parse in partial_parses.

Implement this algorithm in the minibatch_parse function in parser_transitions.py. You can run basic (non-exhaustive) tests by running python parser_transitions.py part_d.

Note: You will need minibatch_parse *to be correctly implemented to evaluate the model you will build in part (e). However, you do not need it to train the model, so you should be able to complete most of part (e) even if* minibatch_parse *is not implemented yet.*

(e) (20 points) We are now going to train a neural network to predict, given the state of the stack, buffer, and dependencies, which transition should be applied next.

First, the model extracts a feature vector representing the current state. We will be using the feature set presented in the original neural dependency parsing paper: *A Fast and Accurate Dependency Parser using Neural Networks*.[8] The function extracting these features has been implemented for you in utils/parser_utils.py. This feature vector consists of a list of tokens (e.g., the last word in the stack, first word in the buffer, dependent of the second-to-last word in

[8] Chen and Manning, 2014, https://nlp.stanford.edu/pubs/emnlp2014-depparser.pdf

the stack if there is one, etc.). They can be represented as a list of integers $\mathbf{w} = [w_1, w_2, \ldots, w_m]$ where m is the number of features and each $0 \leq w_i < |V|$ is the index of a token in the vocabulary ($|V|$ is the vocabulary size). Then our network looks up an embedding for each word and concatenates them into a single input vector:

$$\mathbf{x} = [\mathbf{E}_{w_1}, ..., \mathbf{E}_{w_m}] \in \mathbb{R}^{dm}$$

where $\mathbf{E} \in \mathbb{R}^{|V| \times d}$ is an embedding matrix with each row $\mathbf{E}_w$ as the vector for a particular word w with dimension d. We then compute our prediction as:

$$\begin{aligned} \mathbf{h} &= \mathrm{ReLU}(\mathbf{xW} + \mathbf{b}_1) \\ \mathbf{l} &= \mathbf{hU} + \mathbf{b}_2 \\ \hat{\mathbf{y}} &= \mathrm{softmax}(l) \end{aligned}$$

where $\mathbf{h}$ is referred to as the hidden layer, $\mathbf{l}$ is referred to as the logits, $\hat{\mathbf{y}}$ is referred to as the predictions, and $\mathrm{ReLU}(z) = \max(z, 0))$. We will train the model to minimize cross-entropy loss:

$$J(\theta) = CE(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{j=1}^{3} \mathbf{y}_j \log \hat{\mathbf{y}}_j$$

where $\mathbf{y}_j$ denotes the jth element of $\mathbf{y}$. To compute the loss for the training set, we average this $J(\theta)$ across all training examples.

i. Compute the derivative of $\mathbf{h} = \mathrm{ReLU}(\mathbf{xW} + \mathbf{b}_1)$ with respect to $\mathbf{x}$. For simplicity, you only need to show the derivative $\frac{\partial h_i}{\partial x_j}$ for some index i and j. You may ignore the case where the derivative is not defined at 0.

ii. Recall in part 1b, we computed the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$. Likewise, please compute the partial derivative of $J(\theta)$ with respect to the ith entry of $\mathbf{l}$, which is denoted as $\mathbf{l}_i$. Specifically, compute $\frac{\partial CE(\mathbf{y}, \hat{\mathbf{y}})}{\partial \mathbf{l}_i}$, assuming that $\mathbf{l} \in \mathbb{R}^3$, $\hat{\mathbf{y}} \in \mathbb{R}^3$, $\mathbf{y} \in \mathbb{R}^3$, and the true label is c.
Hints: You may recall from part 1a, $\frac{\partial CE(\mathbf{y}, \hat{\mathbf{y}})}{\partial \mathbf{l}_i} = \sum_j \frac{\partial CE(\mathbf{y}, \hat{\mathbf{y}})}{\partial \hat{\mathbf{y}}_j} \frac{\partial \hat{\mathbf{y}}_j}{\partial \mathbf{l}_i}$, and $\frac{\partial CE(\mathbf{y}, \hat{\mathbf{y}})}{\partial \hat{\mathbf{y}}_j} = 0$ if $j \neq c$.

iii. We will use UAS score as our evaluation metric. UAS refers to Unlabeled Attachment Score, which is computed as the ratio between number of correctly predicted dependencies and the number of total dependencies despite of the relations (our model doesn't predict this).

In parser_model.py you will find skeleton code to implement this simple neural network using PyTorch. Complete the __init__, embedding_lookup and forward functions to implement the model. Then complete the train_for_epoch and train functions within the run.py file.
Finally execute python run.py to train your model and compute predictions on test data from Penn Treebank (annotated with Universal Dependencies).
Note:

- For this assignment, you are asked to implement Linear layer and Embedding layer. Please **DO NOT** use **torch.nn.Linear** or **torch.nn.Embedding** module in your code, otherwise you will receive deductions for this problem.

- Please follow the naming requirements in our TODO if there are any, e.g. if there are explicit requirements about variable names you have to follow them in order to receive full credits. You are free to declare other variable names if not explicitly required.

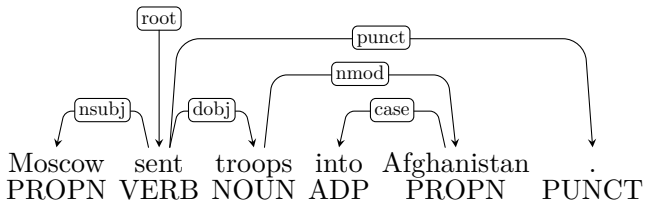
Hints:

- Each of the variables you are asked to declare (self.embed_to_hidden_weight, self.embed_to_hidden_bias, self.hidden_to_logits_weight, self.hidden_to_logits_bias) corresponds to one of the variables above ($\mathbf{W}$, $\mathbf{b}_1$, $\mathbf{U}$, $\mathbf{b}_2$).
- It may help to work backwards in the algorithm (start from $\hat{\mathbf{y}}$) and keep track of the matrix/vector sizes.
- Once you have implemented embedding_lookup (e) or forward (f) you can call python parser_model.py with flag -e or -f or both to run sanity checks with each function. These sanity checks are fairly basic and passing them doesn't mean your code is bug free.
- When debugging, you can add a debug flag: python run.py -d. This will cause the code to run over a small subset of the data, so that training the model won't take as long. Make sure to remove the -d flag to run the full model once you are done debugging.
- When running with debug mode, you should be able to get a loss smaller than 0.2 and a UAS larger than 65 on the dev set (although in rare cases your results may be lower, there is some randomness when training).
- It should take up to **15 minutes** to train the model on the entire training dataset, i.e., when debug mode is disabled.
- When debug mode is disabled, you should be able to get a loss smaller than 0.08 on the train set and an Unlabeled Attachment Score larger than 87 on the dev set. For comparison, the model in the original neural dependency parsing paper gets 92.5 UAS. If you want, you can tweak the hyperparameters for your model (hidden layer size, hyperparameters for Adam, number of epochs, etc.) to improve the performance (but you are not required to do so).

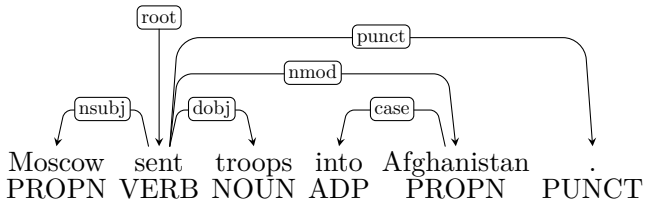
Deliverables:

- Working implementation of the transition mechanics that the neural dependency parser uses in parser_transitions.py.
- Working implementation of minibatch dependency parsing in parser_transitions.py.
- Working implementation of the neural dependency parser in parser_model.py. (We'll look at and run this code for grading).
- Working implementation of the functions for training in run.py. (We'll look at and run this code for grading).
- **Report the best UAS your model achieves on the dev set and the UAS it achieves on the test set in your written submission**.

(f) (12 points) We'd like to look at example dependency parses and understand where parsers like ours might be wrong. For example, in this sentence:



the dependency of the phrase *into Afghanistan* is wrong, because the phrase should modify *sent* (as in *sent into Afghanistan*) not *troops* (because *troops into Afghanistan* doesn't make sense, unless there are somehow weirdly some troops that stan Afghanistan). Here is the correct parse:



More generally, here are four types of parsing error:

- **Prepositional Phrase Attachment Error**: In the example above, the phrase *into Afghanistan* is a prepositional phrase[9]. A Prepositional Phrase Attachment Error is when a prepositional phrase is attached to the wrong head word (in this example, *troops* is the wrong head word and *sent* is the correct head word). More examples of prepositional phrases include *with a rock*, *before midnight* and *under the carpet*.
- **Verb Phrase Attachment Error**: In the sentence *Leaving the store unattended, I went outside to watch the parade*, the phrase *leaving the store unattended* is a verb phrase[10]. A Verb Phrase Attachment Error is when a verb phrase is attached to the wrong head word (in this example, the correct head word is *went*).
- **Modifier Attachment Error**: In the sentence *I am extremely short*, the adverb *extremely* is a modifier of the adjective *short*. A Modifier Attachment Error is when a modifier is attached to the wrong head word (in this example, the correct head word is *short*).
- **Coordination Attachment Error**: In the sentence *Would you like brown rice or garlic naan?*, the phrases *brown rice* and *garlic naan* are both conjuncts and the word *or* is the coordinating conjunction. The second conjunct (here *garlic naan*) should be attached to the first conjunct (here *brown rice*). A Coordination Attachment Error is when the second conjunct is attached to the wrong head word (in this example, the correct head word is *rice*). Other coordinating conjunctions include *and*, *but* and *so*.

In this question are four sentences with dependency parses obtained from a parser. Each sentence has one error type, and there is one example of each of the four types above. For each sentence, state the type of error, the incorrect dependency, and the correct dependency. While each sentence should have a unique error type, there may be multiple possible correct dependencies for some of the sentences. To demonstrate: for the example above, you would write:

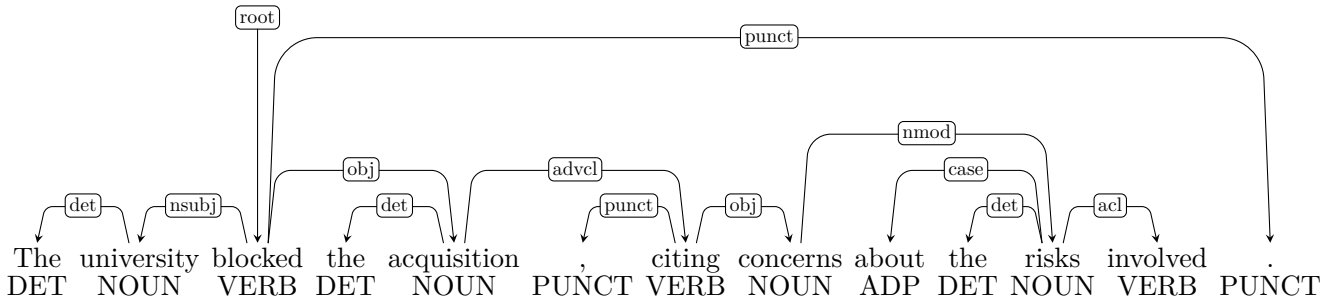
- **Error type**: Prepositional Phrase Attachment Error

[9] For examples of prepositional phrases, see: https://www.grammarly.com/blog/prepositional-phrase/
[10] For examples of verb phrases, see: https://examples.yourdictionary.com/verb-phrase-examples.html

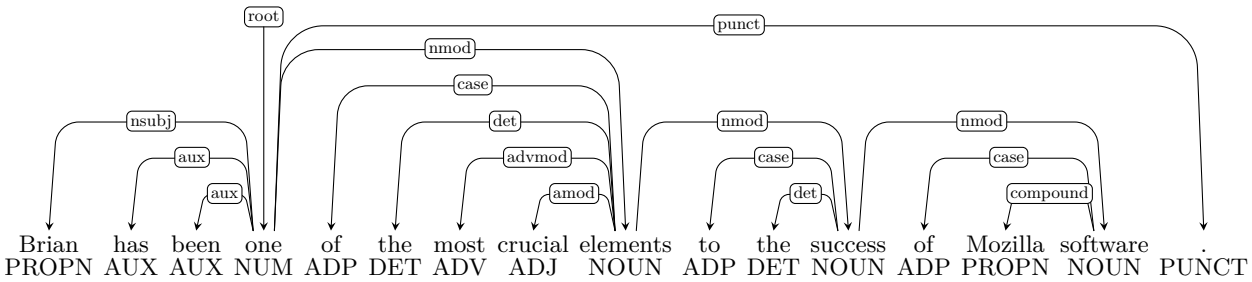
- **Incorrect dependency**: troops → Afghanistan
- **Correct dependency**: sent → Afghanistan

***Note**: There are lots of details and conventions for dependency annotation. If you want to learn more about them, you can look at the UD website:* http://universaldependencies.org[11] *or the short introductory slides at:* http://people.cs.georgetown.edu/nschneid/p/UD-for-English.pdf. *Note that you **do not** need to know all these details in order to do this question. In each of these cases, we are asking about the attachment of phrases and it should be sufficient to see if they are modifying the correct head. In particular, you **do not** need to look at the labels on the the dependency edges – it suffices to just look at the edges themselves.*

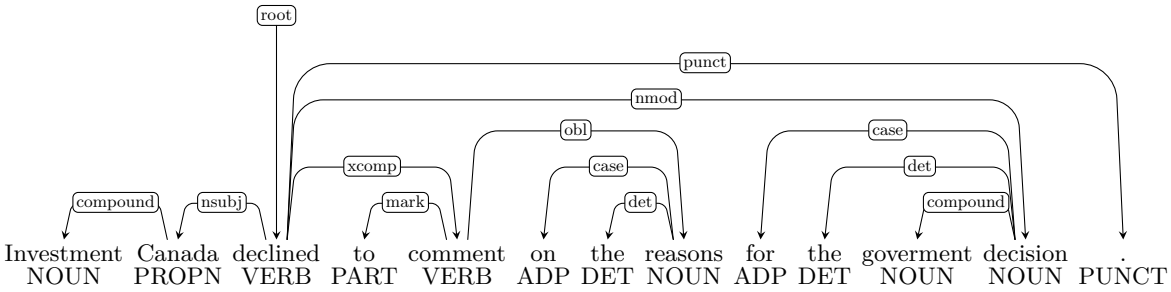
i.



ii.

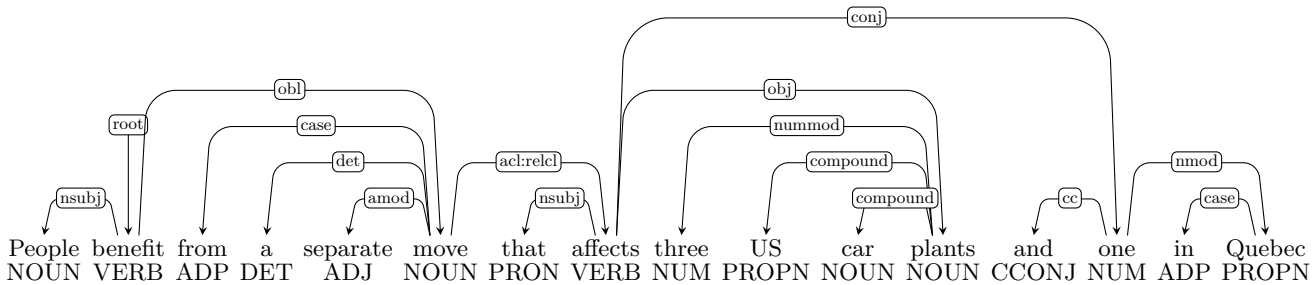


iii.



[11] But note that in the assignment we are actually using UDv1, see: http://universaldependencies.org/docsv1/

iv.



(g) (2 points) Recall in part (e), the parser uses features which includes words and their part-of-speech (POS) tags. Explain the benefit of using part-of-speech tags as features in the parser?

Submission Instructions

You shall submit this assignment on GradeScope as two submissions – one for "Assignment 2 [coding]" and another for 'Assignment 2 [written]":

1. Run the collect_submission.sh script to produce your assignment2.zip file.
2. Upload your assignment2.zip file to GradeScope to "Assignment 2 [coding]".
3. Upload your written solutions to GradeScope to "Assignment 2 [written]".